
A Robot Policy Designed as a Weight-Space Artifact

Exact Tensor Structure in a Capable VLA

Rational conversion, reduced-Gram analysis, and exact weight access

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Tensor decomposition rewrites a multidimensional table as a small set of low-rank
2 factors. In a neural network, those factors can expose which combinations of inputs
3 and outputs are encoded by the weights. Conventional nonlinear operations make
4 such a table difficult to obtain directly. We introduce χ -VLA, a rational vision-
5 language-action model family with a 19.2M convolutional policy and a 20.1M ViT
6 policy, each containing only bilinear, linear, and rational learned operations. Its
7 checkpoint therefore specifies an explicit rational tensor program rather than only
8 an opaque executable function. A mechanical audit and local reconstructions verify
9 that claim. We derive an exact weight-based decomposition of each softmax-free
10 attention layer and reconstruct all 12 modules in three checkpoints with worst
11 relative error 2.56×10^{-7} . An $O(d^2)$ reduced Gram avoids materializing the
12 coefficient tensor. All 36 primary block-head tests favor the rank-128 weight-
13 derived subspace over fixed random controls. The full appendix screen finds the
14 effect in 83/96 pairs and identifies where the signal concentrates across depth. A
15 convolutional instantiation of the same rational model family reaches $93.7\% \pm 0.8\%$
16 on LIBERO-Object, and its three-checkpoint ensemble reaches 96.2%. The three
17 released convolutional checkpoint files associated with those historical aggregates
18 also pass an independent 4,608-case joint-attention certificate with worst relative
19 error 6.36×10^{-16} . We therefore establish a capable model family with exact
20 layerwise structure derived from trained coefficients. We do not yet provide an
21 exact end-to-end decomposition of the complete policy.

22 1 Introduction

23 Neural checkpoints are usually treated as opaque containers for functions. We can run them and
24 collect their activations, but the weights alone rarely provide a readable map from input features to
25 outputs. A post-hoc probe may reveal that an activation correlates with behavior, yet it does not show
26 how the weights construct that feature.

27 A tensor is a multidimensional table, and tensor decomposition is the analogue of low-rank matrix
28 factorization for such tables. When a network uses only additions, multiplications, and ratios
29 of polynomials, its weights can be expanded into coefficient tensors whose entries record input
30 interactions. Decomposition ranks simpler factors inside those coefficients. The intended benefit
31 is an inspectable model artifact: candidate structure comes from the trained parameters. Softmax,
32 ordinary activations, and square-root normalization obstruct this direct conversion in conventional
33 networks. Bilinear language and vision models suggest that changing the primitives can recover it
34 [3, 4]. Whether the resulting artifact can still control a robot is unresolved.

35 We study two linked questions:

36 *Can the learned weights of a specialist VLA specify an explicit rational tensor*
 37 *program, and can exact weight analysis coexist with strong closed-loop control?*

38 Our contributions are:

- 39 1. We construct and mechanically certify a complete VLA from tensor-compatible primitives.
 40 A rational normalization supplies input-specific scaling without leaving the rational function
 41 class.
- 42 2. We derive exact weight-based structure through each softmax-free attention layer and an
 43 $O(d^2)$ reduced-Gram calculation that avoids materializing its high-order coefficient tensor.
- 44 3. We test exact attention subspaces against fixed equal-rank random controls. The weight-
 45 derived subspaces win all 36 primary block-head comparisons.
- 46 4. We establish capability in a convolutional instantiation of the same rational family, reaching
 47 $93.7\% \pm 0.8\%$ over three LIBERO-Object seeds. A three-checkpoint ensemble reaches
 48 96.2% at three times the inference cost.

49 This paper does not claim current state of the art. It also does not claim that comparable mechanisms
 50 are impossible to obtain from conventional policies. The empirical question is whether the algebraic
 51 policy makes exact weight structure directly accessible.

52 2 A rational tensor robot policy

53 2.1 Tensor-compatible primitives

54 Each primitive replaces a familiar transformer component while keeping explicit coefficients. Lin-
 55 ear maps contribute first-order terms, while multiplying learned projections creates higher-order
 56 interactions that remain determined by the weights.

57 **Homogeneous coordinate.** We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative
 58 layer to represent bias, linear, and quadratic terms in one contraction.

59 **Bilinear feed-forward block.** For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

60 The inner sum is a table indexed by one output and two input coordinates. The three matrices give its
 61 CP factorization, the tensor analogue of a low-rank matrix factorization.

62 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

63 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key,
 64 value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with
 65 interaction coefficients determined by the weights.

66 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 67 root. Let $m(x) = d^{-1} \sum_{j=1}^d x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We
 68 deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \quad \text{RNorm}(x) = \frac{x}{\sqrt{s_0}} r(v). \quad (3)$$

69 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
 70 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
 71 rescaling without leaving the rational tensor representation.

72 2.2 Architecture and training

73 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
 74 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
 75 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
 76 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
 77 continuous action dimensions. The complete model has 20,137,352 learned parameters.

78 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
 79 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
 80 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
 81 that verify every requested state was loaded.

82 2.3 Mechanical certificate on the 20.1M ViT checkpoint

83 Architecture definitions can differ from deployed code. We therefore audit the real 20.1M ViT
 84 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

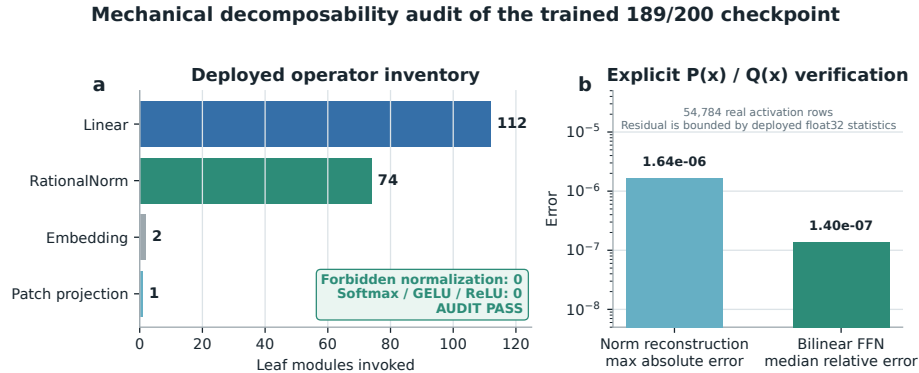


Figure 1: **Mechanical audit of the 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

85 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
 86 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
 87 module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does
 88 not materialize one compact polynomial for the full eight-layer transformer.

89 3 The artifact is a capable closed-loop policy

90 3.1 Evaluation protocol

91 We evaluate all ten LIBERO-Object tasks with:

- 92 • 50 canonical trials per task
- 93 • a 280-step cap
- 94 • three independently trained seeds per architecture
- 95 • 500 rollouts per seed and 1,500 rollouts per architecture

96 Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not
 97 rerun in our codebase, so differences may still include implementation and training choices.

98 3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	19.2M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 19.2\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

99 The three historical convolutional aggregate records report 94.0%, 94.4%, and 92.8%. Their narrow
100 spread is descriptive bounded evidence across independently trained checkpoints.

101 Separately, the paired ensemble evaluation re-evaluates the three members at 93.6%, 95.0%, and
102 93.2%, for a 93.9% mean. Elementwise prediction averaging reaches 96.2%. This is 2.3 points
103 above that paired mean and 1.2 points above its strongest member. The gain is largest on tomato
104 sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one
105 19.2M-parameter policy.

106 3.3 What this result does and does not establish

107 The result shows that the rational architecture supports high closed-loop specialist capability. It is an
108 existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization
109 or state of the art.

110 4 Exact weight-derived structure through attention

111 4.1 Exact layerwise construction

112 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
113 a coefficient table whose entries measure how combinations of query, key, and value coordinates
114 contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-
115 token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to
116 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

117 The deployed attention also normalizes each query and key with Equation 3. Each normalization
118 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
119 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
120 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
121 10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three
122 trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the
123 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
124 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

125 We separately certify the three released convolutional checkpoint files associated with the historical
126 capability aggregates. Their vision stacks contain no attention, so we independently rebuild all eight
127 joint modules and all twelve heads per module from the saved weights. Sixteen deterministically
128 selected cache inputs per checkpoint yield 384 module cases and 4,608 head-input cases. Every
129 output is finite and the worst relative ℓ_2 error is 6.36×10^{-16} , far below the frozen 10^{-6} gate. This
130 same-checkpoint certificate covers the joint-attention equation at each normalized layer input. It does
131 not rebuild the convolutional stem, feed-forward blocks, residual composition, final normalization, or
132 action head as one end-to-end contraction.

133 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
134 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
135 rank important directions without constructing the full table. It matches brute-force materialization to
136 2.13×10^{-14} at toy scale.

4.2 Trained policy result

For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached activations are used only afterward to score how well it preserves the layer’s computation.

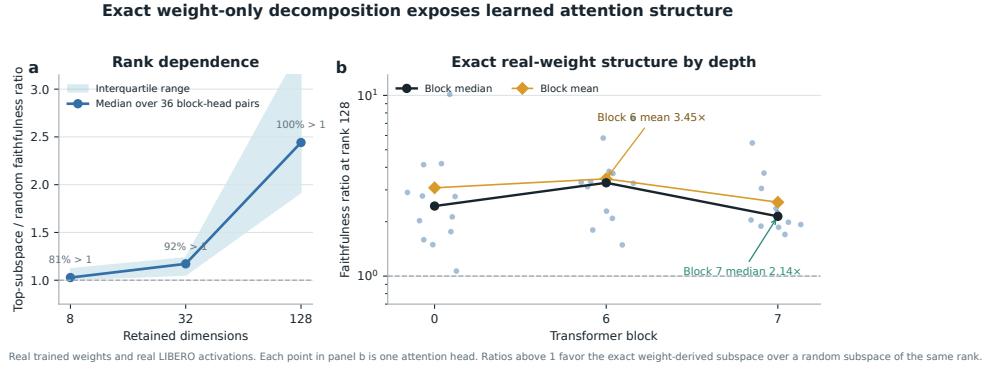


Figure 2: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Related work and limitations

Related work. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a policy designed as a rational tensor program. Tensor and polynomial networks provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success can coexist with weak language use [9–11].

Limitations and conclusion. The benchmark is one specialist suite, so it does not establish broad robotic generalization. Exact reconstruction is certified layer by layer and does not yet produce one compact symbolic contraction of the full policy. The released convolutional files support the joint-attention equation certificate and are associated with the historical capability aggregates. The reduced-Gram subspace screen uses separate ViT checkpoints. Evaluation uses fixed task instructions and does not establish robust instruction grounding. Within that scope, rational restrictions support strong LIBERO-Object control and exact trained-weight certificates. We do not claim a direct coefficient-edit interface.

Reproducibility. The final artifact will provide code, configurations, checkpoint hashes, exact certificates, and per-episode logs. Historical Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to official identifiers before defining task splits. A source-level audit compared all 271,996 cached frames across Object, Spatial, Goal, and LIBERO-10 with their pinned LeRobot revisions. Task identifiers,

170 language joins, states, actions, and resized images matched exactly for every frame. Canonical-state
171 checks fail loudly when requested states are unavailable.

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197 A Supporting results and decisions

198 A.1 Per-task ensemble behavior

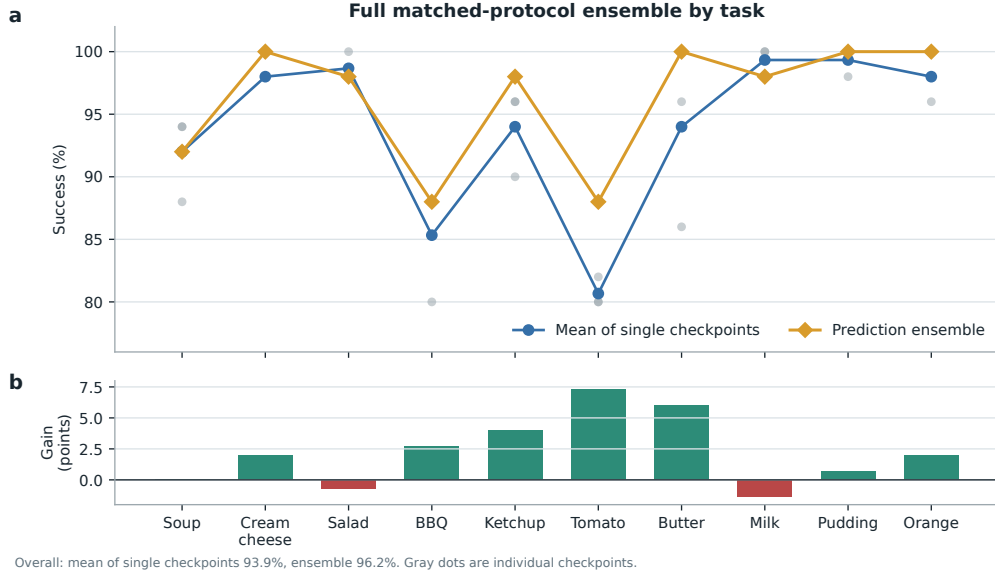


Figure 3: **Per-task effect of prediction averaging in the paired ensemble evaluation.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 19.2M-parameter policy.

199 A.2 Expanded attention screen

200 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 201 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 202 is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96
 203 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider
 204 tested rank.

205 A.3 Numerical scope of rational conversion

206 The deployed degree- $[2/2]$ least-squares rational fit is exact as a rational computation even though it
 207 approximates RMS normalization. Its numerator has coefficients (5.32997, 7.53552, 0.327274) and
 208 the denominator has coefficients (1, 9.64827, 2.60664). Every denominator coefficient is positive,
 209 which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances,
 210 each invoked once per forward pass, on 4,096 deterministic inputs for each of three convolutional
 211 checkpoints. Across 544,542,720 activation rows, the minimum observed denominator is 1.0016 and
 212 no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that
 213 recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined
 214 as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a frozen 10^{-3}
 215 gate.

216 The certificate also bounds what the operator approximates. Observed normalized mean-square ratios
 217 v span 1.66×10^{-4} to 17.17, outside the nominal fit interval $[0.1, 10]$. Across site-checkpoint pairs,
 218 the minimum coverages are 60.018% in that interval and 60.022% with rational-versus-exact-RMS
 219 scale error at most 3.4%. The frozen 99% secondary coverage gates therefore do not pass. The
 220 positive claim is therefore global absence of nonnegative denominator poles and cache-conditioned
 221 floating-point fidelity to the deployed rational operator, not uniform agreement with exact RMS
 222 scaling. Matched alternative-normalizer runtime overhead remains unmeasured.

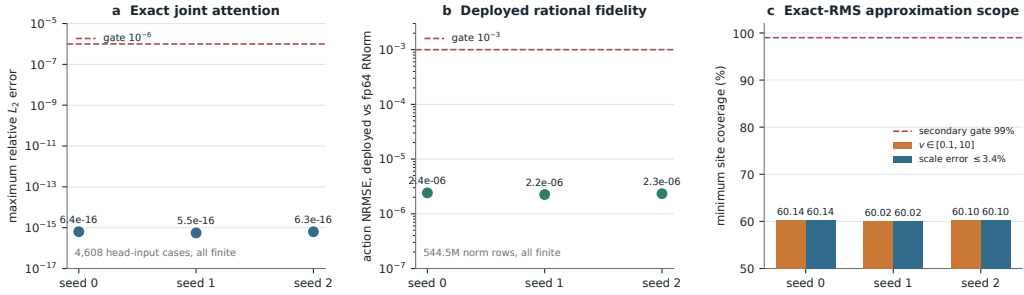


Figure 4: **Frozen structural certificates on the released convolutional files associated with the historical capability aggregates.** Panels (a) and (b) pass their primary gates with large margins. Panel (a) reconstructs all 4,608 audited joint-attention head-input cases. Panel (b) compares deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64. Panel (c) reports the separately frozen exact-RMS approximation coverage, which does not pass its 99% secondary gate.